

Lindsay Cheng

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EDUCATION

University of Toronto St. George

BASc Computer Engineering + PEY Co-op | 3.90/4.0 cGPA

Awards & Honors: First Year Summer Research Fellowship (2025)

Toronto, ON

Expected April 2030

EXPERIENCE

University of Toronto Aerospace Team (UTAT) - Space Systems

Sep 2025 – Present

Software Developer, Attitude Determination & Control Systems (ADCS)

Toronto, ON

- Developing **Python** control system for a 3-axis Helmholtz cage used in satellite **hardware-in-the-loop** testing
- Designing **closed-loop PID controller** with real-time magnetometer feedback to generate arbitrary 3D magnetic field vectors within **2% accuracy**
- Building **automated data logging** pipeline recording field vectors, coil currents, and timestamps at ≥ 1 Hz with CSV export for post-test analysis

Green Technologies Lab (University of Toronto)

May 2025 – Aug 2025

Materials Science Engineering Research Intern

Toronto, ON

- Researched **Phase Change Materials (PCM)** for EV battery applications using novel biochar composites
- Developed **calcium-catalyzed pyrolysis** process achieving **10% faster thermal response** at **200°C lower** processing temperature than industry standard processes
- Automated **Raman spectroscopy** analysis using **Python**, processing 100+ biochar samples and reducing manual analysis time by **80%**

PROJECTS

ClearMark | Swift, Node.js, Express, PostgreSQL, Google OAuth, Gemini API | [Link](#)

- Built a full AI grading and course management iOS app in **Swift**, using **VisionKit** for document scanning and engineering LLM context prompts for rubric-aligned evaluation
- Designed **MVVM** architecture and **PostgreSQL** schema, implementing strict prompt guardrails to extract structured **JSON**, achieving a **98% parsing success rate** and reducing hallucinations
- Deployed **Node.js/Express** backend on Render with **Google OAuth 2.0**, **RESTful APIs**, and **Cloudflare R2** to support role-based access control for instructors and students

Water Bottle Defect Detection System | Python, YOLOv8, OpenCV, SQLite, Tkinter | [Link](#)

- Built automated quality control system using **YOLOv8 + OpenCV**, with a **Tkinter** monitoring dashboard
- Collected and annotated **300+ images** across 4 defect classes to train and validate a **YOLOv8n** model over 100 epochs, achieving **95.9% mAP@50-95**
- Used **ByteTrack** to implement per-bottle ID assignment and multi-object tracking with **SQLite** defect logging

Reversi Game Bot | C | [Link](#)

- Implemented **minimax** engine with **alpha-beta pruning** in **C**, achieving 800ms move decisions at **7-ply depth**
- Designed heuristic evaluation with **position weighting** and **mobility analysis**, improving win rate by **20%** over the baseline model

Send | Swift, YOLOv8, OpenCV, FastAPI, MongoDB | [Link](#)

- Built social bouldering iOS app integrating **YOLOv8** hold detection and pathfinding algorithm, processing wall images in under **2 seconds**
- Deployed **FastAPI** backend with **ArUco marker** calibration for accurate camera-to-wall coordinate mapping

SKILLS

Languages: Python, C/C++, Swift, JavaScript, TypeScript, HTML, CSS

Frameworks & Libraries: React, Nextjs, Nodejs, Express, FastAPI, PyTorch, OpenCV, Drizzle ORM

Technologies: PostgreSQL, MongoDB, SQLite, Redis, Docker, BullMQ, NGINX

Tools & Platforms: Git, GitHub, GitHub Actions (CI/CD), Postman, Turborepo, Cloudflare R2, Linux

Concepts: Version Control, Data Structures, Microservices, SDLC, User Requirements Analysis

Focus Areas: Full Stack, Mobile Development, Computer Vision, Machine Learning, Quality Assurance